

System Architecture & Technical Proposal

Project Name: Custom Multi-Outlet Restaurant Management Platform (Petpooja Integration)

Document Type: Formal Technical & Operational Proposal

Target Execution Framework: Phase 1 (Single Outlet Validation) \rightarrow Phase 2 (Multi-Outlet Scaling)

Executive Summary

This proposal outlines the design, architecture, and deployment strategy for a custom, lightweight operational management system built specifically for restaurant operators.

While Petpooja serves as an efficient front-line Point of Sale (POS) and billing engine, off-the-shelf Enterprise Resource Planning (ERP) software is often overly complex, slow, and expensive. This custom platform acts as a centralized Command Tower. It extracts live transactional data from Petpooja via secure Application Programming Interfaces (APIs) and transforms it into actionable insights for raw material procurement, waste reduction, staff accountability, and multi-location oversight.

Key Operational Challenges Addressed

Information Fragmentation: Eliminates the need to manually log into isolated POS terminals or sift through static spreadsheets to evaluate business performance.

Food Cost Leakage: Resolves the discrepancy between items sold at the billing counter and raw inventory consumed in the kitchen.

Operational Inconsistency: Replaces loose verbal instructions and unstructured messaging groups (e.g., WhatsApp) with standardized digital operational logs.

Scalability Bottlenecks: Establishes a modular infrastructure during the single-outlet phase that can seamlessly onboard additional outlets (up to 5+ locations) without requiring architectural redesigns.

System Architecture & Data Flow

Front-Line Layer (Petpooja POS): Handles counter billing, KOT (Kitchen Order Ticket) printing, and payment processing.

Integration Layer (Webhooks/REST API): Pushes transactional data instantly to the custom system whenever an order is billed, voided, or discounted.

Central Intelligence Layer (Custom Platform): Processes raw POS data, computes inventory degradation based on standardized recipes, and serves analytics to mobile and desktop interfaces.

Functional Modules & Capabilities

Module 1: Owner Control Tower (Real-Time Analytics)

Live Revenue Stream: Aggregated gross sales, net revenue, tax breakdown, and payment mode distribution updated in real time.

Loss Prevention Alerts: Instant notifications triggered upon threshold breaches (e.g., bill cancellations, heavy manual discounts, cash drawer openings without an associated sale).

Item Velocity Tracking: Real-time metrics highlighting top-performing dishes and slow-moving inventory.

Module 2: Recipe-Based Inventory & Stock Variance Engine

Standardized Bill of Materials (BOM): Link menu items directly to raw ingredient usage formulas (e.g., $1 \text{ Paneer Butter Masala} = 200 \text{g Paneer} + 50 \text{g Butter}$).

Automated Stock Depletion: Automatically deducts raw materials from system inventory in real time as sales occur at the POS counter.

End-of-Day Variance Calculation: Compares theoretical stock remaining against physical closing stock entered by the store manager to isolate unexplained inventory loss

Module 3: Digital Operations & Audit Framework

Time-Stamped Opening/Closing Logs: Digital checklists covering equipment status, refrigeration temperature logs, and hygiene inspections.

Visual Verification: Mandatory photo upload requirement for critical station checks prior to shift commencement and closure.